



Corsair unveiled a 128 GB version of their ultra fast Voyager GT flash drives

Intel Pinetrail

The next generation Atom successor will have both memory controller and graphics core on-die

HOW WE TESTED

The CPUs were divided into three categories:

1. Up to Rs. 5,000
2. Rs. 5,001 to Rs. 12,500
3. Rs. 12,501 and above

The first category represents entry-level processors. The second category also represents value, especially the lower rungs. The top end of this middle category has good value for money as most of these CPUs are pretty powerful and should suffice for even power users. The top category consists of the fastest processors and

represents both buyers who want sterling performance for a solid price and those who demand the best, regardless of cost.

Since CPU features or specifications are of very little consequence and performance and price are the only two things that matter, we have ignored features and gave a full weighting (i.e. 100) to performance. Therefore the performance score is the same as the grand total score in our tables.

We used Windows Vista Ultimate 32-bit with SP1. We carefully chose the best, most stable and high-end motherboards to ensure no CPU was bottlenecked. ASUS' GeForce GTX 285 was the graphics card of choice. We used Corsair Dominator memory all round and the timings were carefully kept constant throughout the test.

When testing CPUs we ensure they run at their default stated clock speeds at all times. For this we disable any sort of

thermal throttling down, halt states, temperature monitoring and variable multipliers. We thereby eliminate any chances of performance being affected due to the time lag between speeding up and down, irrespective of either company's claim to the effectiveness of their variable clocking mechanisms. For the Core i7 CPUs we kept hyper-threading enabled, although we did disable it on a subsequent run of the Intel Core i7 975 EE just to see how this affected the scores.

Test Configurations used:

	Intel Core 2 Duo	Intel Core i7	AMD AM3	AMD AM2
Motherboard	ASUS P5Q3 Deluxe	ASUS P6T Deluxe v2	Gigabyte 790FX-UD5	ASUS 780G
RAM	2 x 2 GB DDR3 @ 1333 MHz (8-8-8-20)	3 x 1 GB DDR3 @ 1333 MHz (8-8-8-20)	2 x 2 GB DDR3 @ 1333 MHz (8-8-8-20)	2 x 2 GB DDR2 @ 1066 MHz (5-5-5-15)
Graphics Card	ASUS GeForce GTX 285	ASUS GeForce GTX 285	ASUS GeForce GTX 285	ASUS GeForce GTX 285
Hard Drive	Seagate 7200.11 500 GB	Seagate 7200.11 500 GB	Seagate 7200.11 500 GB	Seagate 7200.11 500 GB

A brief description of the benchmarks used:

Synthetic	Description	Real World	Description
3D Mark Vantage	We only ran the CPU tests. The CPUs FP unit is stressed with a number of physics intensive scenes. Since Vantage scales with multi-core CPUs as well as clock speed it is sensitive to minute changes - a good benchmark	Maya 2008	Multi-core 3D rendering benchmark. Can be set to utilise a single or more cores. Doesn't scale beyond 8 logical cores. We set it to render a scene at 1024 x 1024 pixels, at production quality. Multi-core and single core performance were tested
Wprime 2.0	We recorded the time taken to calculate the square roots of the first 32-million integers. Pure FP stress benchmark and one used commonly in number crunching comparative tests. Scales excellently with multiple cores	WinRAR 3.9 Beta	Test compression/decompression capability
CineBench R10	Developed by Maxon, a 3D scene is used to render a photo-realistic image. Stresses CPU capabilities for features including reflection, ambient occlusion, area lighting and procedural shaders. Uses up to 16 cores. Tests SSE capabilities and has support for SSE 4.2	DivX 7 encoding	Encoding .VOB to DivX. Supports SSE 4.1
Valve Multicore CPU benchmark	Uses the half life 2 engine and simulates particle physics. A good benchmark for judging physics and rendering capabilities of a CPU	Handbrake	Encoding a .VOB file to x264. Supports SSE 4 and supports up to 8 processor cores. Has better multicore support than DivX 7 but is slower
POV Ray 3.62	Raytracing benchmark supports most features like multiple and different light sources, radiosity, surface patterns. Doesn't support SMP and multiple cores at the moment, therefore is a good test of per core performance and therefore CPU architectural performance	Company Of Heroes Tales Of Valor	An RTS game that utilises multiple threads, therefore suitable for multi-core CPUs
ScienceMark 2.0	Membench tests overall processor to memory bandwidth and also the L1 and L2 cache speed and latency. Cryptography uses the AES encryption algorithm to assess the performance of FP math co-processor	Crysis	CPU benchmark was run. This benchmark has heavy emphasis on physics and a lot of destructive environments. A good measure of a processors ability at running modern games. We ran it at 1024 x 768 pixels and low quality to prevent our graphics solution from bottlenecking scores
		Doom 3	The in-game timedemo was run three times and an average of the 2nd and 3rd score were taken. The Doom 3 engine is old but because of its ultra sensitivity to any change in physical hardware is a good indicator of performance. It scales with clock speed primarily, but also scales with the number of cores